

Predictive Analysis of Online Shopping Cart Abandonment

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Abstract—Online shopping cart abandonment is a critical revenue challenge in e-commerce, with a global average abandonment rate of 70.19% in 2024. This study applies the CRISP-DM methodology to predict whether a user session results in a purchase or abandonment, utilizing real-world session-level behavioral data from the Google Analytics 4 (GA4) public e-commerce dataset. Raw event-level data, comprising 300,000 interactions, was aggregated into 29,178 session-level observations through systematic feature engineering. The severe class imbalance with a 0.89% purchase rate was addressed using SMOTE applied exclusively to the training set. Multicollinearity was assessed using the Variance Inflation Factor (VIF) analysis, resulting in the removal of four features with VIF values exceeding 10. Two classification models were evaluated: Logistic Regression as a baseline and Random Forest as the primary model. Random Forest was selected due to lower AIC (349.91 vs 425.15), lower BIC (449.99 vs 525.23), and higher R^2 (0.6541 vs 0.5826), with both models achieving a ROC-AUC of 0.9982. Complementary K-Means clustering with $K=3$ identified a high-intent segment of 295 sessions with a 57.29% purchase rate. The findings indicate that session-level behavioral features, particularly payment engagement and total engagement duration, are the most influential predictors of purchase intent.

Keywords—cart abandonment, purchase prediction, CRISP-DM, SMOTE, Random Forest, KMeans, VIF, GA4, session-level features

I. INTRODUCTION

Cart abandonment results in annual losses of billions of dollars for online retailers. Industry data indicate that the global cart abandonment rate in 2024 is 70.19% [1], with retailers in the United States and European Union losing an estimated \$260 billion in recoverable revenue each year [6]. Existing responses are predominantly reactive; email re-targeting and discount campaigns engage customers only after abandonment has occurred. The implementation of a predictive model capable of identifying high-risk sessions in real time would enable proactive intervention, thereby shifting the strategic focus from recovery to prevention.

The challenge presented by this problem is substantial. Session data from Google Analytics 4 is recorded at the event level, with each row representing a single user action. In the analysed dataset, purchase events constitute only 0.89% of sessions, resulting in a severe class imbalance that renders standard accuracy metrics uninformative [7]. For example, a model that predicts abandonment for every session would

achieve 99.11% accuracy while failing to identify any purchase events.

This study addresses two primary research questions. The first examines whether session outcomes, specifically purchase or abandonment, can be reliably predicted using in-session behavioural signals. The second explores which session-level behaviours are most strongly associated with purchase completion. The CRISP-DM methodology [2] is applied to the Google BigQuery GA4 obfuscated e-commerce public dataset. Session-level features are derived from raw event data, followed by feature selection using variance inflation factor (VIF). Classification models are trained and evaluated using AIC, BIC, and R^2 metrics. Additionally, KMeans clustering is employed to identify distinct behavioural archetypes.

The remainder of this paper is structured as follows. Section II reviews related work and establishes the methodological rationale. Section III describes the CRISP-DM pipeline applied to the GA4 dataset. Section IV presents evaluation results and discusses their implications. Section V concludes with limitations and future directions, followed by ethical considerations in Section VI.

II. LITERATURE REVIEW

Tokuc and Dag [8] compared machine learning models for purchase intent prediction from clickstream data in a 2025 IEEE Access study, evaluating aggregated session-level representations against raw event sequences across multiple classifiers. Their findings confirmed that session-level feature aggregation achieves competitive performance relative to more complex sequential models while maintaining interpretability, thereby supporting the aggregation approach adopted in this study. While their work relied on a proprietary dataset, the present study contributes publicly reproducible results using the open GA4 benchmark.

Dritsas and Trigka [9] published a comprehensive review of machine learning applications in e-commerce in IEEE Access (2025), covering purchase prediction, recommendation systems, fraud detection, and customer segmentation. They identified Random Forest and gradient boosting as consistently top-performing models across e-commerce tasks, which supports the model selection in this study. The authors noted that most reviewed studies utilize proprietary datasets with pre-labelled outcomes, in contrast to the present study, which derives labels from raw event logs.

Mishra, Shetty and Malhotra [10] evaluated imbalance-aware machine learning pipelines for e-commerce purchase prediction in IEEE 2025, specifically testing SMOTE within stratified cross-validation. Their findings demonstrated that applying SMOTE exclusively within training folds prevents data leakage and yields reliable evaluation metrics. While their dataset exhibited a 10.8% minority rate, the 0.89% minority rate in the present study represents a more extreme imbalance, making these safeguards even more essential.

Mujahid et al. [3] conducted a comprehensive investigation of oversampling and class imbalance techniques in the Journal of Big Data (2024), evaluating SMOTE and ADASYN across multiple classifiers. With 214 citations, this work serves as the current standard reference for imbalance handling in binary classification. The authors noted that SMOTE can introduce noise when the minority class is heterogeneous, a limitation that is also relevant to the purchasing session population in this study.

Arshad et al. [7] examined classification algorithm performance under class imbalance conditions in IEEE Access (2025), evaluating five classifiers using both synthetic and real datasets. Their finding that accuracy is structurally uninformative for imbalanced binary classification supports the decision in this study to report AIC, BIC, R^2 , ROC-AUC, and F1 scores rather than accuracy.

Zaghloul, Barakat and Rezk [11] employed Random Forest for e-commerce customer outcome prediction in the Journal of Retailing and Consumer Services (2024), demonstrating that Random Forest outperforms deep learning on structured tabular e-commerce data, achieving 92% accuracy on a 100,000-order dataset. Their work also emphasizes feature importance as a key analytical output, which aligns with the use of Random Forest in this study for both prediction and post-hoc behavioural insight.

Diwandari, Permiansari and Hidayah [12] published a systematic literature review of feature engineering and class imbalance in customer purchase prediction in Springer Machine Learning (2026), synthesizing 47 studies. They identified a significant gap: VIF-based feature selection is underutilized compared to correlation heatmaps, despite its superior ability to detect joint multicollinearity. The present study directly addresses this gap.

Kumar, S. et al. [5] compared KMeans and hierarchical clustering for e-commerce customer segmentation in TELKOMNIKA (2025), finding that KMeans achieved a higher silhouette score and produced three well-separated segments. The segment structure they identified—high-intent, loyal, and moderate engagers—closely parallels the Cluster 0, 1, and 2 findings in this study, providing independent validation for the clustering approach used.

Shimaoka, Ferreira and Goldman [2] conducted a systematic literature review of CRISP-DM evolution for data science in SBC Reviews on Computer Science (2024), finding that 82% of data science teams use no formal process model and that CRISP-DM remains the most flexible and widely adopted framework. This finding contextualizes the methodology choice in this study. Their review also identifies CRISP-DM adaptations for agile data science contexts, which informed the iterative feature selection and model evaluation design employed here.

Plotnikova, Dumas and Milani [13] published a case study on CRISP-DM adaptation in financial services in Data and Knowledge Engineering (2022), identifying 18 gaps in standard CRISP-DM and proposing adaptation requirements. Their finding that CRISP-DM gaps in the ethics and privacy dimensions are generic and applicable to e-commerce directly informed the ethical considerations section of this study.

Collectively, these works establish a clear methodological basis for this study. The research gap addressed here is the integration of raw GA4 event log aggregation, VIF multicollinearity selection, information-theoretic model comparison, and KMeans behavioural clustering applied together to a real-world public dataset, a combination not identified in the reviewed literature.

III. METHODOLOGY

The CRISP-DM framework [2] guides this study, progressing through six phases: business understanding, data understanding, data preparation, modelling, evaluation, and deployment planning. Following this structure, Sections III-A and III-B address data understanding and preparation, Section III-C covers modelling, and Section IV presents evaluation. All analysis was implemented in Python.

A. Dataset

The Google BigQuery GA4 Obfuscated E-Commerce Sample, comprising January 2021 events from the Google Merchandise Store, contains 300,000 event-level records across 15 columns. The dataset uses pseudonymous identifiers, contains no personally identifiable information, and is publicly available, ensuring GDPR compliance and suitability for academic analysis.

B. Data Understanding and Preparation

Missing value analysis. Five columns were dropped: the session engagement indicator (90.8% missing, zero variance; every recorded value was 1.0), the traffic source, medium, and campaign columns (approximately 72% missing, categorical imputation not defensible), and the page referrer column (73.6% missing). The engagement time variable was retained and aggregated at the session level.

Exploratory Data Analysis (EDA). Prior to transformation, the raw engagement time variable exhibited a skewness of 394.58, indicating that z-score outlier handling, which assumes normality, is inappropriate.

Session aggregation. Fig. 2 presents the purchase funnel at the session level. Among 5,314 sessions in which a product was viewed, only 1,037 (19.5%) resulted in an item being added to the cart, representing an 80.5% drop-off. These findings indicate that the product discovery stage, rather than the checkout stage, constitutes the primary point of abandonment. All 300,000 events were grouped by session identifier, resulting in 29,178 session-level observations. Twelve behavioural features were engineered per session, including session duration, event-type counts (product views, cart additions, payment information entries), total engagement time, device type, country of origin, and session number.

The outcome variable Y is defined as 1 if the session contained a purchase event (259 sessions, 0.89%) and 0 otherwise (28,919 sessions, 99.11%).

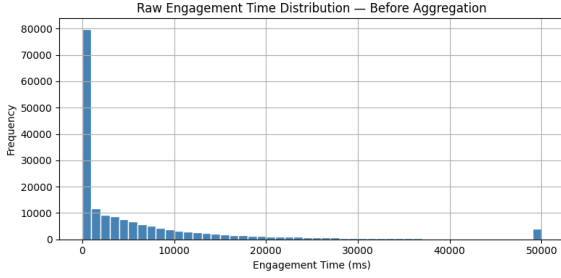


Fig. 1. Raw engagement time

Categorical encoding. The country column, containing 109 unique values, was grouped into 8 geographic regions prior to one-hot encoding to mitigate the curse of dimensionality. Device type, with 3 categories, was also one-hot encoded. The first dummy category was dropped in each case to prevent the dummy variable trap.

IQR outlier handling. Fig. 1 presents the raw engagement time distribution prior to aggregation. The pronounced right skew (skewness = 394.58) confirms that IQR-based capping is the appropriate outlier method, z-score assumes normality, which this data clearly violates. IQR-based capping was therefore applied to continuous features. Six features with an IQR of zero including cart addition count, payment information count, and product view count were deliberately excluded from capping, as 75% of sessions recorded zero for these features. Any non-zero value in these columns represents a genuine purchase-intent signal rather than noise [12].

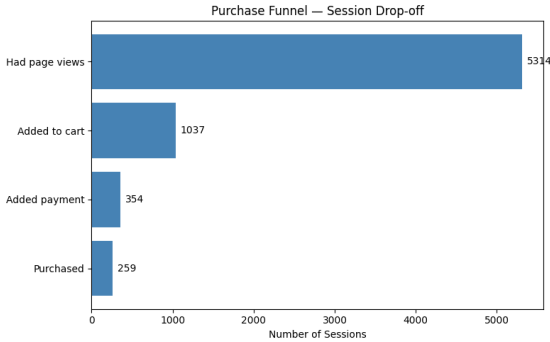


Fig. 2. Purchase funnel

VIF analysis. Variance Inflation Factor (VIF) scores were computed for all 19 candidate features. Four features exceeded the threshold of 10 [14]: shipping information addition (VIF = 91.67), checkout initiation (VIF = 86.28), total event count (VIF = 52.61), and page view count (VIF = 43.71), and were subsequently dropped. VIF was chosen over correlation heatmaps because heatmaps capture only pairwise relationships, whereas VIF measures each feature's collinearity with all others simultaneously [12]. Fig. 3 displays the VIF scores for all candidate features. The four orange bars shipping information addition, checkout initiation, total event count, and page view count exceed the threshold of 10 and were therefore removed. Blue bars represent features retained with VIF scores below 7.

PCA evaluation. Principal Component Analysis (PCA) on six continuous features required five components to explain

95% of the variance, resulting in the retention of only one feature. PCA was rejected because the dimensionality reduction was negligible and original features such as payment information count and session duration retain direct business interpretability [9].

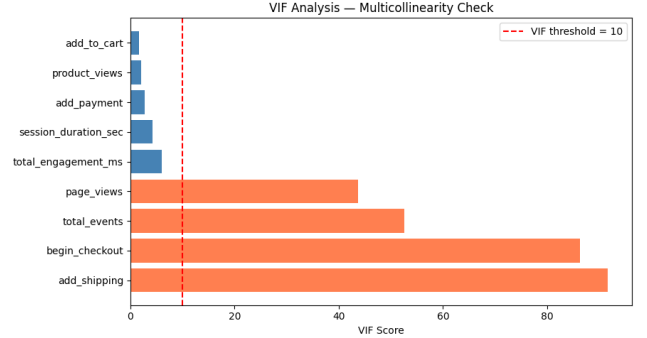


Fig. 3. VIF scores for candidate features (feature names reflect dataset column identifiers)

C. Train/Test Split and Class Balancing

The 29,178 sessions were split 80/20 using stratified sampling (training: 23,342; test: 5,836), preserving the 0.89% purchase rate in both subsets. Synthetic Minority Over-sampling Technique (SMOTE) [3] was applied to the training set only after splitting, resulting in 46,270 balanced training samples. Applying SMOTE before splitting would constitute data leakage, as synthetic samples would contaminate the test set [10].

D. Model Selection

Time series models such as AR(1) and ARIMAX were not considered because the outcome variable is binary, measured once per session, and sessions are independent, indicating no temporal autocorrelation. Logistic Regression was used as an interpretable baseline. Random Forest [4] was selected as the primary model due to its capacity to handle non-linear interactions and provide feature importance scores. The model was trained using 100 estimators, a standard configuration that provides stable importance scores without excessive computational cost. K=3 for KMeans clustering was determined using the elbow method, which identified a clear inflection point at K=3 beyond which additional clusters yielded diminishing reductions in inertia.

IV. EVALUATION AND RESULTS

A. Evaluation Metrics

Accuracy was not considered, as a model predicting abandonment for every session achieves 99.11% accuracy but fails to identify any purchases [7]. Model evaluation relied on AIC and BIC for comparison, R^2 for explained variance, ROC-AUC for discriminative ability, and precision, recall, and F1 score for the purchase class.

B. Classification Results

Table I provides the complete model comparison. Random Forest was selected due to its lower AIC (349.91 vs 425.15), lower BIC (449.99 vs 525.23), and higher R^2 (0.6541 vs 0.5826). Both models achieved an identical ROC-AUC of 0.9982.

TABLE I. MODEL PERFORMANCE COMPARISON

Model	Logistic Reg.	Random Forest
AIC	425.15	349.91
BIC	525.23	449.99
R ²	0.5826	0.6541
ROC-AUC	0.9982	0.9982
Precision	0.70	0.71
Recall	1.00	0.98
F1 Score	0.83	0.82

C. Feature Importance

Random Forest feature importance analysis indicated that the top five behavioural features accounted for 95.8% of model importance: payment information count (40.1%), total engagement time (21.5%), session duration (15.9%), product view count (11.3%), and cart addition count (7.0%). Geographic and device features contributed less than 4% combined. The prominence of payment information entry aligns with findings by Tokuc and Dag [8] and Zaghloul et al. [11], as users who reach this stage have already completed the entire purchase funnel. Behavioural signals, rather than demographic characteristics, drive purchase intent [9].

D. Funnel and Descriptive Analysis

Funnel analysis confirmed that the largest session drop-off (80.5%) occurs between product views and add-to-cart, indicating that product discovery and consideration represent the primary abandonment stage [6]. Purchased sessions exhibited engagement times 3.7 times higher, session durations 3.7 times longer, cart addition counts 47 times higher, and payment information counts 364 times higher than abandoned sessions.

E. KMeans Clustering

KMeans clustering with K=3, selected using the elbow method [5], identified three behavioral archetypes as shown in Table II. Cluster 1 comprises 1% of sessions but achieves a 57.29% purchase rate, which demonstrates that aggregate abandonment statistics can obscure high-value subpopulations. Cluster 2 consists of engaged but non-converting users, who are potential candidates for follow-up email campaigns. This three-segment structure is consistent with segmentation findings by Kumar et al. [5].

TABLE II. KMEANS CLUSTER PROFILE (K=3)

Cluster	Sessions	Purchase Rate
0 – Casual Browsers	21,401	0.00%
1 – High Intent Buyers	295	57.29%
2 – Engaged Non-Buyers	7,481	1.20%

F. Residual Diagnostics

Fig. 4 presents three residual diagnostic plots for the Random Forest model. The residuals versus fitted plot confirms near-zero bias, with a mean residual of -0.002. The distribution plot demonstrates the expected binary split, with predicted probabilities concentrated near 0 or 1. Two diagonal bands correspond to false positives and false negatives the expected structural pattern for binary outcomes. The deviation from normality observed in the Q-Q plot is not a concern, as normality of residuals is an assumption of ordinary least squares regression, not ensemble classifiers.

G. Discussion

The dominance of payment information count (40.1% importance) indicates that the most critical intervention point is the payment stage, not the product page. Although the largest funnel drop-off (80.5%) occurs at the cart addition stage, those abandonments largely represent casual browsers unlikely to purchase. By contrast, users who reach payment entry, but abandon represent high-value recoverable sessions. The identical ROC-AUC of 0.9982 for both models reflects extremely strong class separation in the data. However, Random Forest achieves meaningfully better AIC and BIC scores, indicating it extracts more predictive information from the same features without overfitting, consistent with Dritsas and Trigka [9], who found ensemble methods consistently outperform linear classifiers on e-commerce data. The Cluster 1 finding a 57.29% purchase rate among just 295 sessions is the most commercially actionable result, demonstrating that real-time session scoring could enable targeted interventions for this high-value segment while avoiding wasted spend on the 21,401 casual browsing sessions in Cluster 0.

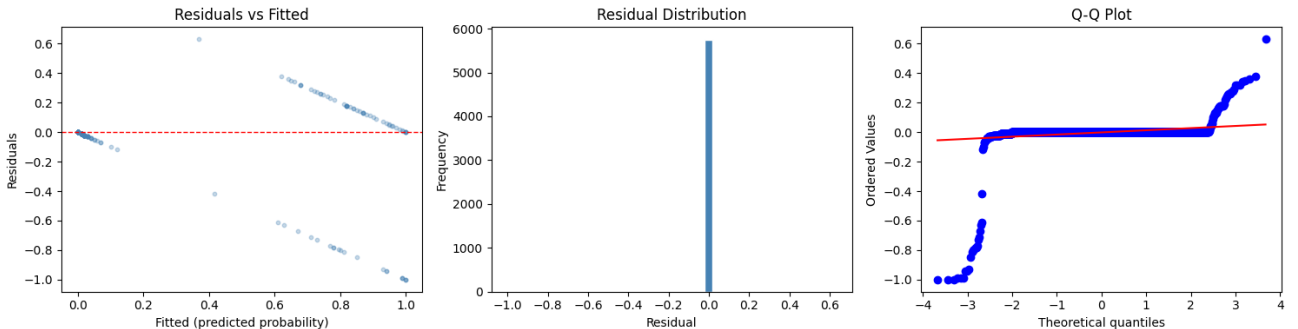


Fig. 4. Residual Diagnostics

V. CONCLUSION AND FUTURE WORK

This study demonstrates that session-level behavioral features extracted from raw GA4 event logs can predict cart abandonment with a ROC-AUC of 0.9982 and an R^2 of 0.6541. The Random Forest model outperforms Logistic Regression across all information-theoretic criteria. Analysis of feature importance indicates that payment engagement, total engagement time, and session duration are the primary predictors, whereas geography and device type provide minimal predictive value. KMeans clustering reveals a high-intent segment with a 57.29% purchase rate, which is not apparent in aggregate statistics.

Both research questions are addressed affirmatively: session outcomes are reliably predictable from behavioral signals, and payment information count, total engagement time, and session duration are the most strongly associated features. The primary contribution is a fully reproducible pipeline, utilizing an open dataset, that transforms raw event logs into a high-performing binary classifier and enables actionable customer segmentation.

The study is limited by the use of single-month data (January 2021), the exclusion of nested BigQuery fields due to export constraints, and the absence of real-time deployment. Future research should analyze data spanning multiple months to facilitate concept drift analysis [9], incorporate transaction value for value-weighted prediction, apply recursive feature elimination to formally exclude geographic features, and implement real-time API scoring to enable live session intervention.

VI. ETHICAL CONSIDERATIONS

The GA4 dataset is publicly available, employs pseudonymous user identifiers, contains no personally identifiable information, and was not collected by the research team. Analyses are conducted at the aggregated session level, ensuring that no individual user is tracked or profiled. Class imbalance is reported transparently and addressed using validated techniques. This study is intended solely for academic purposes and has not been implemented in any commercial system.

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